

Prototype Dashboard & Dataset Input

SmartCity

Insight AI

Project

SDG 11 — Sustainable Cities and Communities

AI Pipeline

01 Information Extraction

02 Complaint Classification

03 Similar Grouping

04 Pattern Detection

05 Root-Cause Analysis

06 Insight & Action Report

SmartCity Insight AI

AI-Powered Urban Grievance Pattern & Root-Cause Analyzer

Identify recurring complaint patterns, explore possible contributing factors, and support evidence-based human investigation.

AI-Assisted

SDG 11

Pattern Analysis

Human Verified

Project Overview

SmartCity Insight AI analyzes synthetic urban grievance data to identify recurring complaint patterns, possible contributing factors, and areas requiring further investigation.

Project objective

Transform a collection of urban complaints into structured insights that can help humans understand recurring patterns.

Important: The system is a decision-support prototype. It does not determine confirmed physical causes and does not make government decisions.

Communities

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Responsible AI

AI suggestions support analysis. Human verification is required before action.

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Data Input

Upload the synthetic grievance dataset to begin the analysis.

Upload Dataset.csv

Drag and drop file here

Limit 200MB per file • CSV

Browse files

Please upload the synthetic Dataset.csv file to view the dashboard.

How the system works

Stage	Purpose
Dataset	Collect complaint records
Information Extraction	Extract useful information
Classification	Identify complaint categories
Grouping	Group similar complaints
Pattern Detection	Detect recurring patterns
Root-Cause Analysis	Suggest possible contributing factors

Important: The system is a decision-support prototype. It does not determine confirmed physical causes and does not make government decisions.

Data Input

Upload the synthetic grievance dataset to begin the analysis.

Upload Dataset.csv

Drag and drop file here

Limit 200MB per file • CSV

Browse files

Dataset.csv 4.9KB

Dataset loaded successfully — 50 complaint records detected.

Executive Dashboard

Total Complaints

50

Categories

7

Verified Patterns

11

Dataset Fields

6

Dashboard values are calculated from the uploaded synthetic dataset and the verified project analysis.

Complaint Category Analysis

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Overview

Dataset

Pattern Analysis

Root-Cause Analysis

Insight Report

Complaint Overview

Distribution of reported complaints across the seven categories present in the synthetic dataset.

Complaint Categories

Drainage

Noise

Public Transport

Road Damage

Streetlight

Waste

Waterlogging

0

1

2

3

4

5

6

7

8

9

10

11

Category Summary

Category	Complaints
Waterlogging	11
Drainage	9
Streetlight	8
Waste	8
Road Damage	7
Public Transport	5
Noise	2

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Dataset Snapshot

Most Reported Category

Highest Category Count

Records Used

Waterlogging

11

50

Complaint frequency represents reported complaints in the synthetic dataset. Frequency alone does not prove actual severity or physical cause.

Overall Summary

What the AI found

Recurring urban complaint patterns

Repeated reports across specific locations

Complaint categories and frequencies

Possible contributing factors for investigation

Evidence IDs connecting patterns to source complaints

What the AI does NOT claim

It does not prove physical causes

It does not establish actual severity

It does not replace human investigation

It does not make government decisions

Synthetic complaint frequency is not the same as real-world impact

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AI Analysis Workflow

Stage	Purpose
1. Dataset	Synthetic urban grievance records
2. Information Extraction	Extract structured information from complaints
3. Complaint Classification	Assign complaint categories
4. Similar Grouping	Group similar complaints
5. Pattern Detection	Identify recurring patterns and locations
6. Root-Cause Analysis	Suggest possible contributing factors
7. Insight & Action Report	Create a structured investigation report
8. Human Verification	Validate findings using real-world evidence

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OverviewDatasetPattern AnalysisRoot-Cause AnalysisInsight Report

Dataset Explorer

Explore the synthetic complaint records used by the analysis.

Rows

50

Columns

6

Unique Complaint IDs

50

Uploaded Records

ID	Complaint	Area	Date	Category	Context
G001	Water is standing near the bus stop after heavy rain.	Zone A	02-Aug-26	Waterlogging	Heavy rainfall
G002	Drain is blocked near the market road.	Zone A	03-Aug-26	Drainage	No rainfall
G003	Road near the bus stop flooded after rain.	Zone A	05-Aug-26	Waterlogging	Heavy rainfall
G004	Streetlight is not working near the community hall.	Zone B	05-Aug-26	Streetlight	Night
G005	Garbage has accumulated near the market.	Zone C	06-Aug-26	Waste	High footfall
G006	Waterlogging happened again near the bus stop.	Zone A	08-Aug-26	Waterlogging	Rainfall
G007	Drain water is overflowing onto the road.	Zone A	09-Aug-26	Drainage	Rainfall

G045	Traffic noise is disturbing the area near the market.	Zone C	16-Sep-26	Noise	Peak traffic
G046	Garbage collection has been delayed near the market again.	Zone C	17-Sep-26	Waste	High footfall
G047	Several streetlights are still not working near the school road.	Zone D	18-Sep-26	Streetlight	Night
G048	Waterlogging is reported near the community hall after heavy rain.	Zone B	19-Sep-26	Waterlogging	Heavy rainfall
G049	Drainage overflow has been reported near the community hall.	Zone B	20-Sep-26	Drainage	Rainfall
G050	The bus service has been delayed during the evening peak.	Zone E	21-Sep-26	Public Transport	Evening peak

Dataset Structure

Field	Purpose
ID	Unique complaint identifier
Complaint_Text	Original complaint description
Area	Reported area/location
Date	Complaint date
Category	Complaint category
Context	Additional contextual information

Pattern & Hotspot Analysis

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Pattern & Hotspot Analysis

Verified complaint groupings created from the synthetic dataset.

The patterns below represent recurring complaint groupings. They should be interpreted as reported patterns, not confirmed real-world conditions.

Verified Complaint Patterns

Pattern	Complaint Count	Evidence IDs
Waterlogging near bus stops	7	G001, G003, G006, G010, G017, G029, G036
Drainage near markets	3	G002, G007, G037
Waste near markets	6	G005, G009, G015, G020, G039, G046
Non-working streetlights	8	G004, G008, G016, G021, G028, G033, G040, G047
Waterlogging near schools	2	G011, G041
Road damage / potholes	7	G013, G014, G019, G026, G031, G038, G043
Public transport issues	5	G022, G023, G034, G044, G050
Waterlogging near community halls	2	G024, G048
Drainage near community halls	2	G025, G049
Waste near schools	2	G027, G032
Traffic noise near markets	2	G035, G045

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Pattern Frequency



Pattern frequency represents reported complaints in the synthetic dataset. It does not prove actual severity or cause.

Evidence Reference

Each pattern is connected to the complaint IDs that support the grouping.

Pattern	Complaint Count	Evidence IDs
Waterlogging near bus stops	7	G001, G003, G006, G010, G017, G029, G036
Drainage near markets	3	G002, G007, G037
Waste near markets	6	G005, G009, G015, G020, G039, G046
Non-working streetlights	8	G004, G008, G016, G021, G028, G033, G040, G047
Waterlogging near schools	2	G011, G041
Road damage / potholes	7	G013, G014, G019, G026, G031, G038, G043
Public transport issues	5	G022, G023, G034, G044, G050
Waterlogging near community halls	2	G024, G048
Drainage near community halls	2	G025, G049
Waste near schools	2	G027, G032

Possible Contributing Factors

AI-assisted analysis suggests possible factors that could be investigated. These are hypotheses, not confirmed causes.

Possible contributing factors must be checked against real-world evidence before any conclusion or action.

Pattern	Possible Contributing Factors	Evidence Strength
Waterlogging near bus stops	Drainage capacity; drainage design; stormwater management; possible blockages	Medium
Drainage near markets	Debris or waste in drains; structural damage; insufficient drainage capacity	Medium
Waste near markets	Collection frequency; available waste-management resources; increased waste generation	Medium
Non-working streetlights	Electrical faults; damaged infrastructure; maintenance issues; power supply problems	Medium
Waterlogging near schools	Drainage capacity; drainage design; stormwater management; possible blockages	Low
Road damage / potholes	Road maintenance; heavy traffic loads; construction quality	Medium
Public transport issues	Scheduling; fleet availability; staffing	Medium
Waterlogging near community halls	Drainage capacity; drainage design; stormwater management; possible blockages	Low
Drainage near community halls	Debris or waste in drains; structural damage; insufficient drainage capacity	Medium
Waste near schools	Collection frequency; available waste-management resources; waste generation during school hours	Medium
Traffic noise near markets	Traffic volume; noise barriers; surrounding urban conditions	Low

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Human Verification

AI-generated contributing factors are hypotheses only. They must be checked against real-world evidence.

- ✓ Check relevant infrastructure and physical conditions.
- ✓ Check maintenance and service records where available.
- ✓ Verify reported blockages, damage, or service issues.
- ✓ Compare AI suggestions with real-world observations.
- ✓ Do not treat possible factors as confirmed causes.

Responsible Interpretation

Observed Evidence

Information directly represented by complaint records and verified complaint groupings.

AI Interpretation

Possible contributing factors generated from observed patterns and requiring human verification.

Insight & Action Report

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Insight & Action Report

A structured summary connecting observed complaint patterns, possible contributing factors, and recommended investigation.

Human officials should review the findings based on real-world evidence, community impact, available resources, and relevant local conditions before deciding on any action.

Issue	Location	Complaint Count	Observed Evidence	Possible Contributing Factor
Waterlogging	Near bus stops	7	Repeated reports of waterlogging near bus stops after rainfall	Drainage capacity, design, stormwater management,
Drainage	Near markets	3	Repeated reports of drainage-related complaints near markets	Debris, structural damage, or insufficient drainage ca
Waste	Near markets	6	Repeated reports of waste accumulation near markets	Collection frequency, resources, or increased waste g
Streetlights	Multiple reported locations	8	Repeated reports of non-working streetlights	Electrical faults, infrastructure damage, maintenanc
Waterlogging	Near schools	2	Repeated reports of waterlogging near schools after rainfall	Drainage capacity, design, stormwater management,
Road Damage	Multiple reported locations	7	Repeated reports of potholes or damaged road surfaces	Road maintenance, traffic loads, or construction qua
Public Transport	Multiple reported locations	5	Reports of late buses, overcrowding, or inconsistent service	Scheduling, fleet availability, or staffing
Waterlogging	Near community halls	2	Repeated reports of waterlogging near community halls after rainfall	Drainage capacity, design, stormwater management,
Drainage	Near community halls	2	Repeated reports of drainage-related complaints near community h	Debris, structural damage, or insufficient drainage ca
Waste	Near schools	2	Repeated reports of waste accumulation near schools	Collection frequency, resources, or waste generation
Traffic Noise	Near markets	2	Repeated reports of traffic noise near markets	Traffic volume, noise barriers, or surrounding urban c

Insight & Action Report

A structured summary connecting observed complaint patterns, possible contributing factors, and recommended investigation.

Human officials should review the findings based on real-world evidence, community impact, available resources, and relevant local conditions before deciding on any action.

Factor	Recommended Investigation	Important Limitation
Drainage capacity, design, stormwater management, or possible blockage, or insufficient drainage capacity	Check drainage capacity, blockages, design, and maintenance records	Possible factors are hypotheses and require real-world verification
Collection frequency, resources, or increased waste generation	Inspect drains, blockages, structural condition, and capacity	Complaint reports do not prove the physical cause
Electrical faults, infrastructure damage, maintenance, or power supply	Review collection schedules, service records, and waste-management resources	Reported waste volume may not represent actual waste generation
Drainage capacity, design, stormwater management, or possible blockage, or insufficient drainage capacity	Inspect affected streetlights and review maintenance and power records	Reports require physical verification
Road maintenance, traffic loads, or construction quality	Inspect drainage systems and check for blockages and capacity issues	Low evidence strength; further evidence is required
Scheduling, fleet availability, or staffing	Inspect road conditions and review maintenance and construction records	Possible contributing factors are not confirmed causes
Drainage capacity, design, stormwater management, or possible blockage, or insufficient drainage capacity	Review schedules, fleet availability, and staffing information	Service complaints require operational verification
Collection frequency, resources, or waste generation during school hours, or surrounding urban conditions	Inspect drainage systems and verify reported blockages	Low evidence strength; further evidence is required
	Inspect drains and check structural condition and capacity	Complaint reports do not prove the physical cause
	Review collection schedules and waste-management resources	Reported complaints do not establish the underlying cause
	Check traffic conditions and, where appropriate, measured noise levels	Low evidence strength; further evidence is required

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Export Report

Download Insight & Action Report

The exported report contains the current verified analysis summary and associated limitations.

Overall Summary

What the AI found

- Recurring urban complaint patterns
- Repeated reports across specific locations
- Complaint categories and frequencies
- Possible contributing factors for investigation
- Evidence IDs connecting patterns to source complaints

What the AI does NOT claim

- It does not prove physical causes
- It does not establish actual severity
- It does not replace human investigation
- It does not make government decisions
- Synthetic complaint frequency is not the same as real-world impact

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Sustainability Context

SDG 11 — Sustainable Cities and Communities

The project explores how AI-assisted analysis can identify recurring urban grievance patterns and support further investigation.

Responsible AI

- Synthetic demonstration data
- No real citizen PII
- Possible factors clearly labelled
- Human verification required
- AI does not make government decisions

Important Project Limitations

SmartCity Insight AI

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Important Project Limitations

1. Synthetic dataset

This prototype uses demonstration data rather than live citizen complaint data.

2. Complaint reporting bias

Complaint volume reflects reported complaints and may not represent actual severity or the complete situation.

3. Possible causes are not confirmed causes

AI-generated contributing factors are hypotheses that require independent verification.

4. Human-in-the-loop

Real-world decisions should be made only after appropriate human review and verification.

5. No automated government decision-making

The system is designed as an analysis and decision-support prototype.